

User guide

How to run OpenBedside, read the interface activity page, change settings, and fix the usual problems. For first-time installation, start with [START-HERE-install-on-your-Mac.md](#).

Three ways to run it

Command	What runs	Use it for
<code>openbedside demo</code>	gateway, pump simulator and a stand-in EHR, all in one window	seeing it work; showing someone
<code>openbedside ehr-sim plus</code> <code>openbedside run</code>	the same, in two windows, so you can watch every HL7 message arrive	testing and development
<code>openbedside run -c your.toml</code>	the gateway with your own adapters and a real (test) interface engine	integration work

Before any of them, in each Terminal window or tab:

```
cd ~
source ~/.venvs/openbedside/bin/activate
```

Stop anything with **Ctrl-C**.

The web page

Open **<http://127.0.0.1:8080>** while the gateway runs. It refreshes every two seconds and has three pages, on the menu at the top: **Activity**, **Devices** and **Patients**.

Who can change things

Anyone who can open the page can look. Registering devices and associating patients needs one of:

- **no admin password set** (the default): only a browser on the same computer as the gateway can make changes. The page says so in a banner;
- **an admin password**: anyone who signs in (button at top right). Set it once:

```
openbedside set-password # or: openbedside set-password -c your.toml
```

Viewers who are not signed in see patient identifiers masked to their last four characters. Keep the page on localhost regardless: there is no encryption yet.

Activity

OpenBedside

Activity

Devices

Patients

this computer: changes allowedv0.2.0

Not a medical device. Not for clinical use. Test data only: use synthetic patient identifiers, this release has no encryption. Auto-programming works against the simulator only.

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STATUS

EHR link: **connected** Queue: sent 2

DEVICE CONNECTIONS (ADAPTERS)

Adapter	Connection	Messages	Skipped	Last error
acme-json (example)	connected	9	0	

DEVICES

Device	Patient	Module	Status	Actual rate	Sources
BIO-004512 ACME (fictional) ACME-2 · ICU 204 1	TEST0042 adt-location	ACME-7731-L	infusing	12 mL/h	primary heparin 12 mL/h (1200 [iU]/h) VTBI 250 mL, remaining 246.8 mL, delivered 3.2 mL
BIO-004512 ACME (fictional) ACME-2 · ICU 204 1	TEST0042 adt-location	ACME-7731-R	infusing	100 mL/h	primary sodium chloride 0.9% 75 mL/h VTBI 500 mL, remaining 500 mL, delivered 0 mL secondary cefazolin 100 mL/h VTBI 50 mL, remaining 23.333 mL, delivered 26.667 mL

EVENTS

When (UTC)	Event	Where	Detail
17:46:26	patient-associated derived	ACME-7731	{ "patient": "TEST0042", "source": "adt-location" }
17:46:23	delivery-start derived	ACME-7731-R	{ "note": "first observation already infusing" }
17:46:23	delivery-start derived	ACME-7731-L	{ "note": "first observation already infusing" }

OUTBOUND TO EHR

#	Control id	Status	Tries	Last result
2	PD2026092617462600002	sent	1	AA
1	PD2026092617462300001	sent	1	AA

ORDERS RECEIVED (PCD-03)

None.

Status. Whether the link to the EHR is up, the last connection error if not, and how many messages are pending, sent or rejected.

Device connections. One row per adapter: connected or not, messages received, messages skipped (could not be translated), and the last error. The first place to look when a device does not appear.

Devices. One row per pump module of every registered device: the identifier sent to the EHR, its location, its patient and where that came from, status, the rate it is actually running, and each source hanging on it (primary and, when running, secondary) with drug, programmed rate, dose rate, VTBI, volume remaining and delivered.

Events. What happened, newest first: delivery start, stop and complete, rate changes, KVO, secondary to primary, communication lost and restored, and the patient decisions (associated, cleared, conflict, review). Every one is labelled *derived*, because OpenBedside works them out

rather than the device reporting them, and says so.

Outbound to EHR. Every message queued for the hospital: its control id, whether it was sent, how many tries it took, and the hospital's reply (AA means accepted).

Orders received. Test orders sent to the simulator and how they were answered.

The same information is available as JSON at <http://127.0.0.1:8080/api/state> for anyone who wants to build their own view.

Devices: registration

Nothing from a device reaches the EHR until it is registered.

Seen but not sent lists every device an adapter can hear that is not registered, is retired, or is registered as a different type (a ventilator registration will not carry pump data). Click **Register** and the form fills in what the device reported.

OpenBedside

Activity

Devices

Patients

this computer: changes allowed

v0.2.0

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SEEN BUT NOT SENT

Devices an adapter can hear that are not registered, are retired, or are registered as a different type. Nothing from them reaches the EHR.

None.

REGISTERED DEVICES

Type	Vendor	Device id	EHR id	Location	Patient from	Status	
Infusion pump	ACME (fictional)	ACME-7731 ACME-2	BIO-004512 asset	ICU 204 1	device, ADT, manual	active	EditRetire

REGISTER A DEVICE

Device type

Infusion pump

Vendor (as the adapter reports it)

ACME (fictional)

Device id (serial, as the device reports it)

ACME-7731

Model

ACME-2

Hospital asset tag

BIO-004512

EUI-64 (16 hex characters, if it has one)

Send to the EHR as (what the nurse scans)

Asset tag

Unit (point of care)

ICU

Room

204

Bed

1

Facility

Connection binding (only for devices that never send an id)

adapter.port

Find this device's patient from

☒ the device (wristband scanned at the device)

☒ ADT: the patient in this device's bed

☒ a person, on the Patients page

Notes

Save

Clear

Saved.

Field	What it is for
Device type	infusion pump, ventilator, other. Must match what the adapter delivers
Vendor, device id	how the device identifies itself. Together they are the key: two vendors can both ship serial 12345
Asset tag, EUI-64	the hospital's tag, and the device's 64-bit id if it has one

Field	What it is for
Send to the EHR as	what the nurse scans to tie the device to a patient: serial, asset tag or EUI-64. If this does not match the label, device data never lands in a chart
Unit, room, bed, facility	where the device lives. Sent in PV1-3, and needed for association by ADT
Find this device's patient from	which sources may name the patient: the device, ADT for this bed, a person
Connection binding	only for devices that never send their own id, such as a serial device behind a terminal server port

Retire stops a device being sent without deleting its history. Identity is never an IP address.

Patients: association

Each registered device's patient is decided from the sources its registration allows:

- **device**: the device reported an identifier (a wristband scanned at the pump);
- **manual**: someone associated them on this page;
- **ADT for this bed**: the hospital's ADT feed has exactly one patient in the device's registered bed.

State	Meaning	What the EHR receives
associated	the allowed sources agree	the patient, and PV1 with their location and visit
none	no source has a patient	PID-3 Unknown
conflict	sources disagree, or two patients in one bed	PID-3 Unknown until fixed
review	ADT alone would change the patient while the device is running	PID-3 Unknown until someone clicks Confirm new patient

OpenBedside
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DEVICE TO PATIENT

Every source allowed for a device must agree, or no patient is sent. A patient change from ADT alone while a device is running waits for a person to confirm.

Device	Location	Association	Patient	Detail
BIO-004512 ACME (fictional)	ICU 204 1	associated	TEST0042 adt-location	from adt-location

ASSOCIATE A DEVICE WITH A PATIENT

Device
Patient identifier (MRN)
Assigning authority

BIO-004512 (ACME (fictional))
default from config
Associate

CENSUS FROM ADT

Patient	Visit	Class	Location	Last event
TEST0042 HOSP	V42	I	ICU 204 1	A01

Recent ADT: A01 ***0042 at ICU 204 1

A manual association stays until it is cleared. Clear it when the device comes off the patient, or it will conflict with the next one.

Census from ADT is what the hospital's ADT feed has told the gateway: identifiers, visit, class and location. Names are never stored.

To try it without a hospital:

```
openbedside send-adt --event A01 --mrn TEST0002 --unit ICU --room 101 --bed A
openbedside send-adt --event A02 --mrn TEST0002 --unit ICU --room 102 --bed B # transfer
openbedside send-adt --event A03 --mrn TEST0002 # discharge
```

Use made-up identifiers only. This release is not for real patient data.

Settings

Every setting, with its default, is in [config/example.toml](#). Copy it somewhere outside iCloud-synced folders if you like, edit the copy, and run `openbedside run -c path/to/copy.toml`. The ones most often changed:

Setting	Default	Change it when
<code>[ehr] host , port</code>	127.0.0.1, 6661	pointing at a real (test) interface engine
<code>[ehr] receiving_app , receiving_facility</code>	EHR, HOSP	the interface engine expects specific MSH-5 and MSH-6 values
<code>[gateway] periodic_seconds</code>	60	the hospital wants snapshots more or less often

Setting	Default	Change it when
[gateway] offline_after_seconds	30	your device reports less often than every 30 seconds
[gateway] database	openbedside.db	you want the queue somewhere specific
[ui] port	8080	something else already uses 8080
[orders] enabled	true	set false with real devices; orders are simulator-only
[registry] unknown_devices	hold	send sends unregistered devices with patient Unknown (lab use only)
[patients] assigning_authority	HOSP	the hospital's code for PID-3.4
[adt] host , port	127.0.0.1, 6663	the interface engine sends ADT somewhere else
[adt] enabled	true	no ADT feed

Files it creates

openbedside.db (plus -wal and -shm files while running) in the folder you start it from. It holds the outbound queue, the event log, the device registry, the ADT census, manual associations and the admin password hash. Delete it to start clean: pending messages, registrations and the password go with it.

Problems and fixes

You see	Cause	Fix
command not found: openbedside	the environment is not active in this window	source ~/.venvs/openbedside/bin/activate
no such file or directory: .../activate	the environment was never created	python3 -m venv ~/.venvs/openbedside , then install again
Browser: unable to connect	the gateway is not running, or stopped with an error	look at the gateway's window
A port is already in use	another copy is running, or another program uses the port	stop the other copy, or change the port in your config
EHR link: not connected	nothing is listening at [ehr] host:port	start openbedside ehr-sim , or check the interface engine address

You see	Cause	Fix
Messages stuck <i>pending</i>	the EHR is down or not acknowledging	fix the link; they send in order once it is back
A message <i>rejected</i>	the receiver replied AR	read "last result"; the queue moves on, the message is kept
Adapter shows skipped messages	your device sent something the adapter could not translate	run <code>openbedside check</code> and read the warnings
AE module is infusing from send-order	correct behaviour	a pump will not take a new program while running
A device is connected but nothing is sent	it is not registered	Devices page, <i>Seen but not sent</i> , Register
Patient shows <i>conflict</i>	the allowed sources name different patients	read the detail; fix the ADT, clear the manual association, or change the device's sources
Patient shows <i>review</i>	the ADT patient in the bed changed while the device was running	check the bedside; Confirm new patient if it is right
"sign in to make changes"	an admin password is set	Sign in, top right
Save or Associate is greyed out	you are viewing from another computer with no password set	set a password, or use the gateway's own computer

Updating

Replace the files in your OpenBedside folder with the new version (or `git pull` if you cloned it), then:

```
source ~/.venvs/openbedside/bin/activate
pip install -e ~/Documents/OpenBedside
```

Windows

Install Python 3.11 or newer from python.org (tick "Add python.exe to PATH"). In PowerShell:

```
py -m venv $HOME\.venvs\openbedside
$HOME\.venvs\openbedside\Scripts\Activate.ps1
pip install -e "$HOME\Documents\OpenBedside[dev]"
openbedside demo
```

Removing it

Delete the environment folder `~/ .venvs/openbedside` , the OpenBedside folder, and any `openbedside.db` files. Nothing else on the computer is changed by installing it.